Project Report: Road Disruption Intelligence Tool
Date: September 9, 2026
Scope: AI-Based Indian Highway Transport Disruption Mapping

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1. Executive Summary
The Road Disruption Intelligence Tool is a fully automated, end-to-end system designed to monitor, extract, and map real-time hig

2. Current Architecture & Tech Stack (Lite Version)
To eliminate heavy infrastructure requirements (like Docker and PostgreSQL), the tool was built using a lightweight, native stack:
* **Database:** SQLite (`roaddb.db`) with native JSON geometry storage.
* **Backend:** Python FastAPI running native background threads for continuous ingestion.
* **Frontend:** React + Vite, utilizing interactive map rendering for `LineString` geometries.
* **AI/NLP Layer:** Local `langchain_ollama` (Llama3) for entirely free, offline natural language extraction.
* **Geocoding:** Free OSM Nominatim API for translating locations to coordinates.

3. Key Features Implemented

A. Automated Multi-Source Data Ingestion
The system features a robust background worker that continuously queries free APIs (Google News RSS) using an expanded arra

B. AI-Powered NLP Extraction
Unstructured news text is fed into a locally running Llama3 model. The AI parses the text and strictly outputs a JSON schema con
* `event_type` (e.g., FLOODING, ACCIDENT, CONSTRUCTION)
* `impact_type` (e.g., FULL_CLOSURE, TRAFFIC_DIVERSION)
* Geographic markers (Cities, States, Landmarks, Highways)
* Temporal Data (Start and End times)

C. Automated Evidence Fusion
When multiple news articles report on the same event, the backend automatically fuses them together. It resolves temporal conflic

D. Interactive Mapping Dashboard
The React frontend dynamically fetches active disruptions and renders them as thick road segments on the map. Clicking a segm

E. Machine Learning Dataset Generation
For every highly verified event (Certainty >= 0.7), the pipeline automatically logs a highly structured JSONL record to `ml_dataset.

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4. Technical Hurdles & Resolutions
1. **Reddit API Blocking:** Attempted to integrate free Reddit JSON scraping, but Reddit's recent unauthenticated bot-blocking (H
2. **Local LLM Batching Bottleneck:** The local Llama3 model takes ~20 seconds per article, causing the UI to appear frozen whi
3. **Timezone Fusion Crashes:** Merging naive SQLite datetimes with timezone-aware NLP datetimes caused pipeline crashes. *

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5. Future Enhancements (Impact of Paid APIs)
Upgrading the current "Lite/Free" architecture with enterprise paid APIs would drastically scale the system's speed, accuracy, and

1. Paid LLM APIs (OpenAI GPT-4o / Google Gemini Pro)
* **Current Issue:** Local Llama3 takes 20-30 seconds per article, limiting ingestion to a few dozen articles an hour. It also occasi
* **Paid Impact:** Cloud LLMs offer sub-second inference. The system could process **thousands of articles per minute** with ne

2. Premium Data Providers (EventRegistry / NewsAPI / X Enterprise API)
* **Current Issue:** Google News RSS truncates results to the "Top 20" articles and hides highly localized, obscure sources.
* **Paid Impact:** Real-time firehose access to every published article and tweet in India. We could ingest live updates from regio

3. Enterprise Routing (Google Maps Directions API)
* **Current Issue:** We currently draw simple bounding boxes and `LineStrings` based on coordinate bounding.
* **Paid Impact:** We could snap the exact disruption coordinates to actual road networks. The UI would perfectly highlight the ex

4. Enterprise Database (PostgreSQL + PostGIS)